Chapter 23

Electric Potential

Chapter Objectives

- Work done on a charged particle in an electric field
- The concept of electric potential
- Potential difference between one point and another

Chapter Outline

1. Electric Potential Energy
2. Electric Potential
3. Calculating Electric Potential
4. Equipotential Surfaces
5. Potential Gradient

23.1 Electric Potential Energy

- **Potential Energy** U is the work when a conservative force acts on a particle to move it from one point to another.

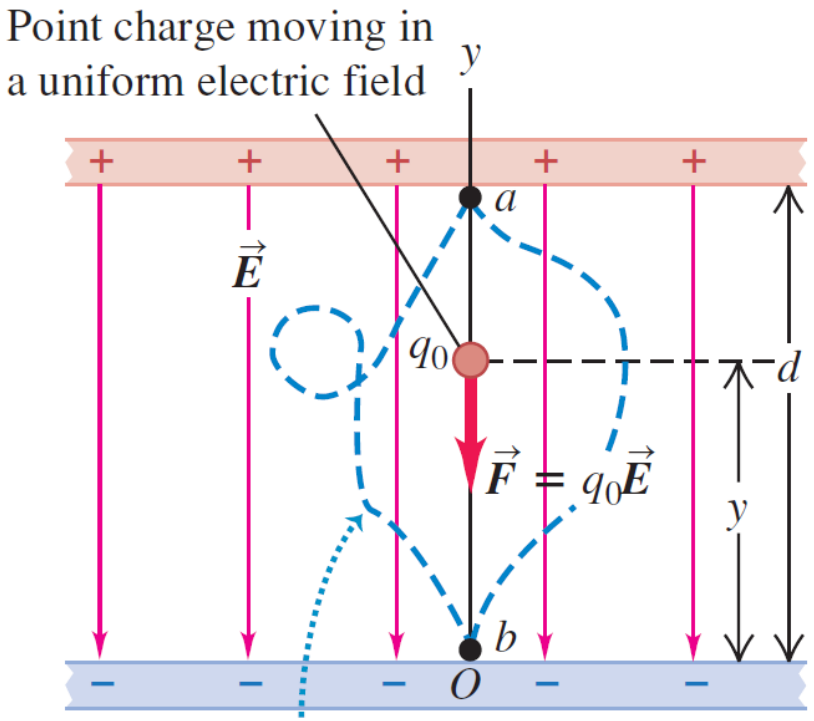
$$W_{a\to b} = U_a - U_b = -(U_b - U_a) = -\Delta U \qquad (23.2)$$

(work done by a conservative force)

Electric Potential Energy in a Uniform Field

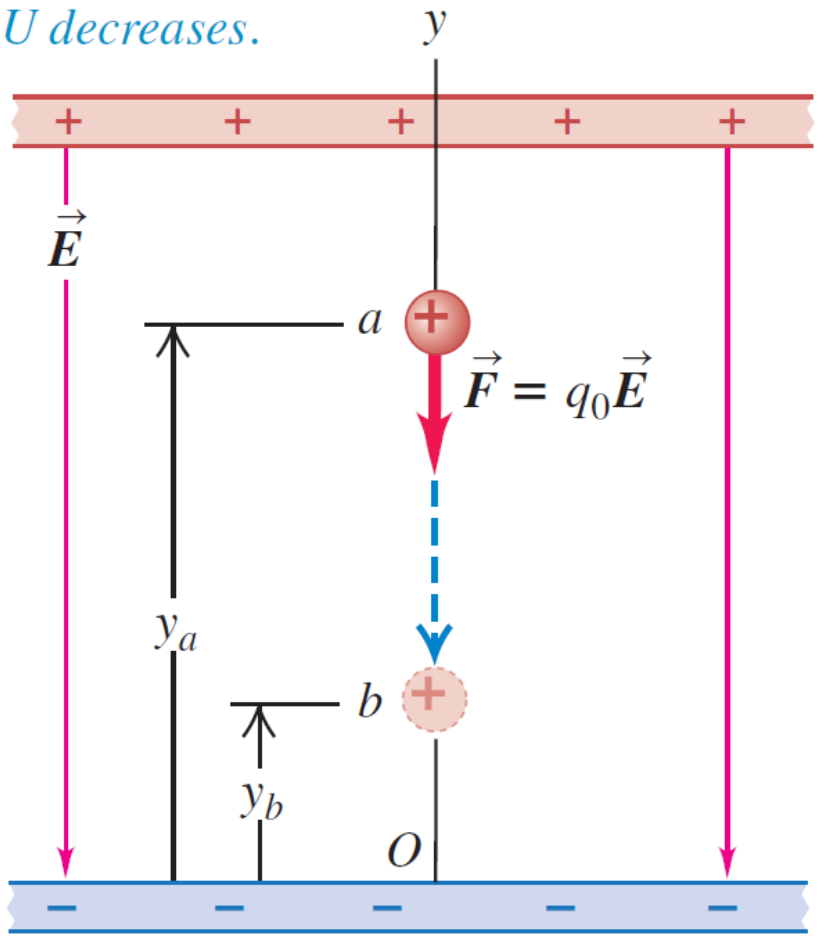
- The force on a test charge that moves from one point to another is constant and independent of its location.
- The work done by this force is also independent of the particle's path.

$$W_{a\to b} = -(U_b - U_a) = qE(y_a - y_b)$$



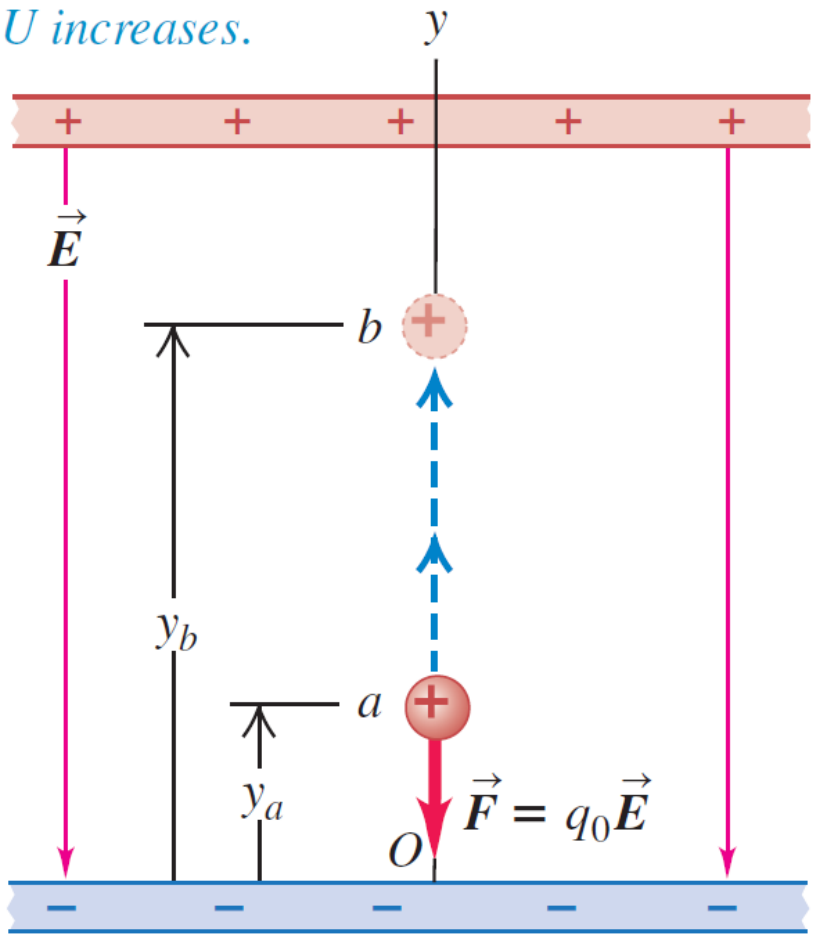
(a) Positive charge moves in the direction of $\vec{E}$:

- Field does *positive* work on charge.
- *U decreases.*



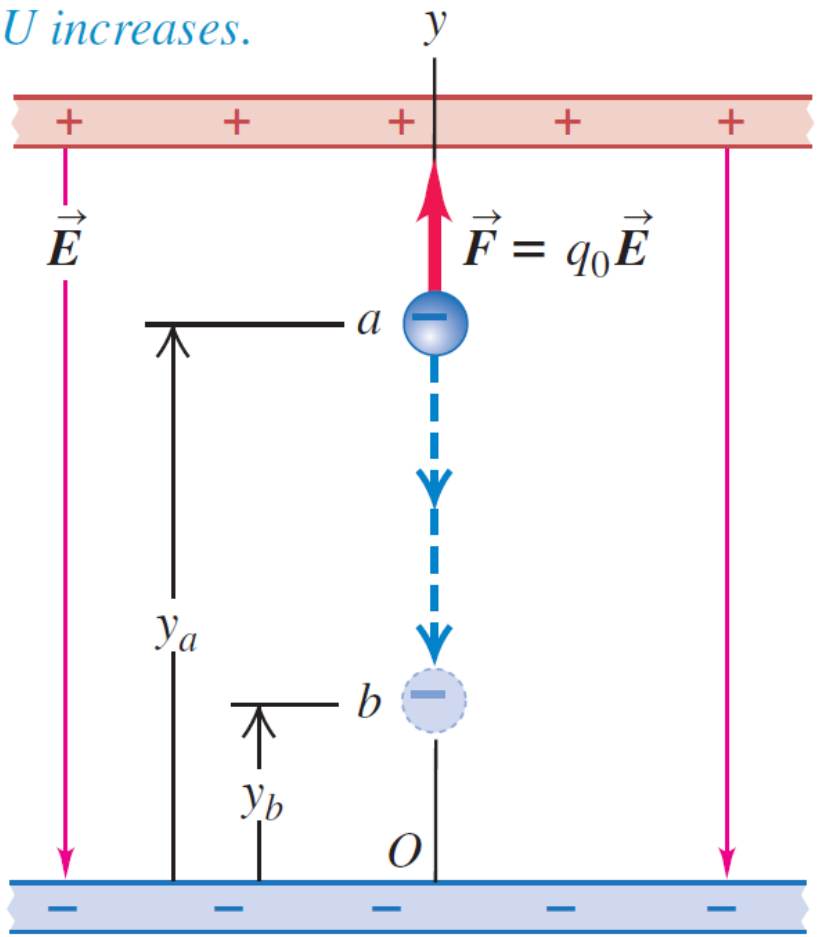
(b) Positive charge moves opposite $\vec{E}$:

- Field does *negative* work on charge.
- *U increases.*



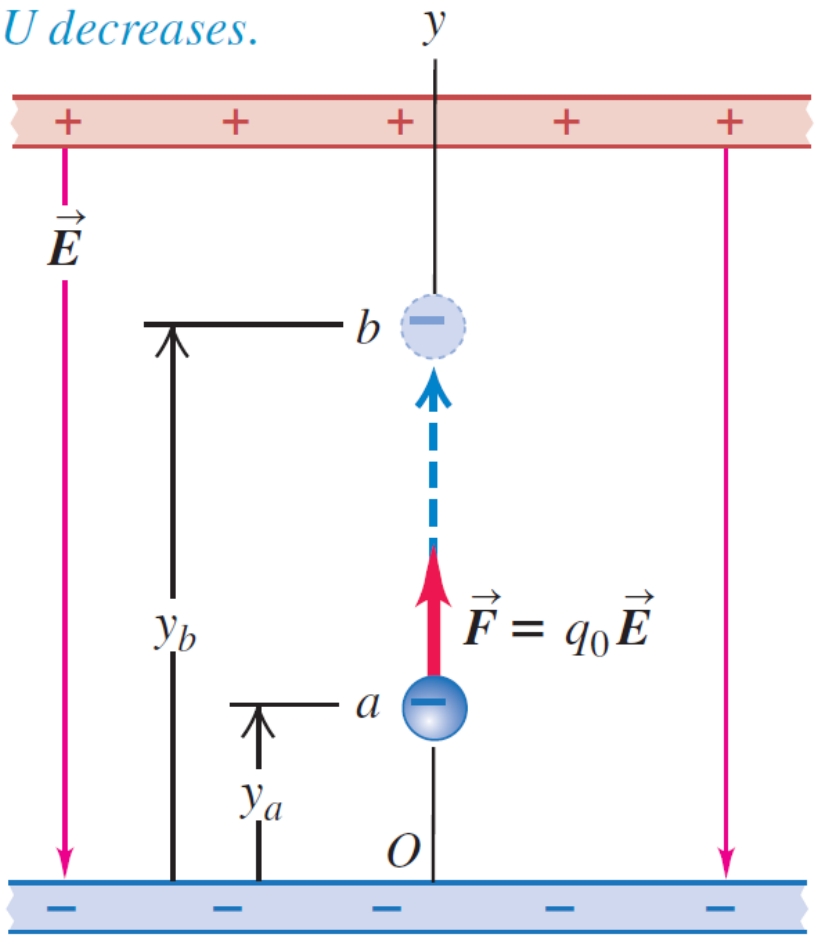
(a) Negative charge moves in the direction of $\vec{E}$:

- Field does *negative* work on charge.
- *U increases.*



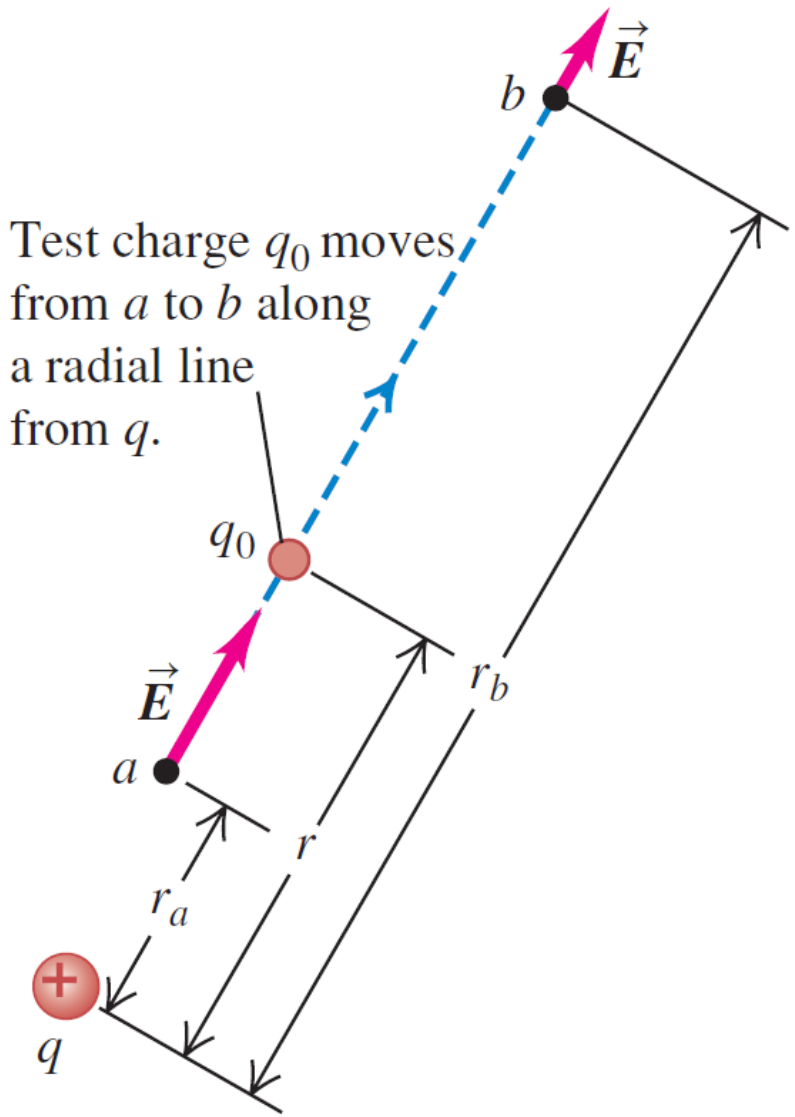
(b) Negative charge moves opposite $\vec{E}$:

- Field does *positive* work on charge.
- *U decreases.*



Electric Potential Energy in a Uniform Field

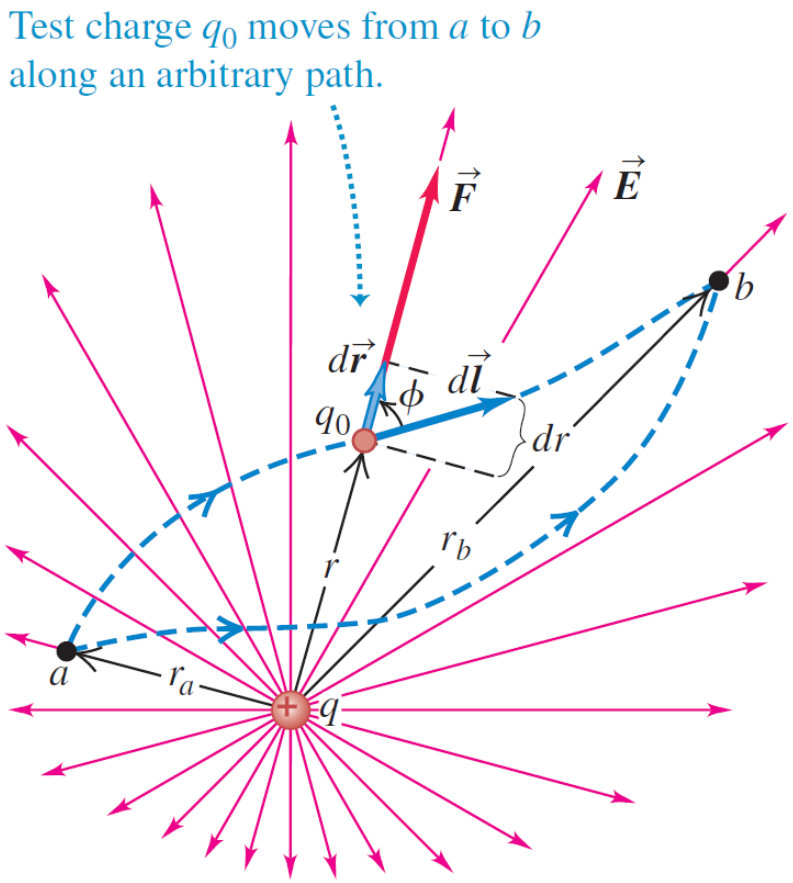
- The idea of electric potential energy isn't restricted to the special case of a uniform electric field.
- We can apply this concept to a point charge in any electric field caused by a static charge distribution.
- This is shown in the figure in the next slide, where displacement is along a radial line in the electric field set up by a single, stationary point charge.


Test charge q_0 moves from a to b along a radial line from q.
$\vec{E}$
b
q_0
$\vec{E}$
a
r_b
r
r_a
q

23.1 Electric Potential Energy

- In fact, the work done by the electric force in moving the test charge from points a to b is the same for all possible paths from a to b.

- This is shown in the figure on the right side where a and b do not lie on the same radial line and the test charge takes a non-linear path.



23.1 Electric Potential Energy

- The potential energy U when the test charge q_0 is at any distance r from charge q is thus

$$U = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r} \qquad (23.9)$$

(electric potential energy of two point charges q and q_0)

- Potential energy is always defined relative to some reference point where $U = 0$.

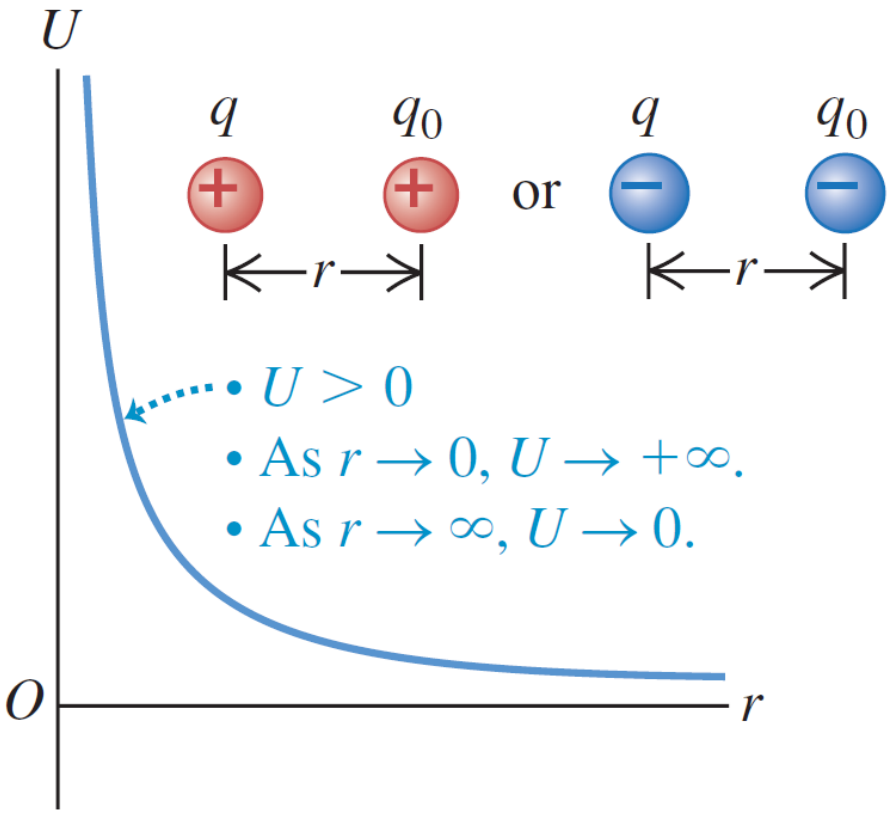
23.1 Electric Potential Energy

- It is a shared property of the two charges q and q_0 because it is a consequence of the interactions (attractive or repulsive) between these two bodies.

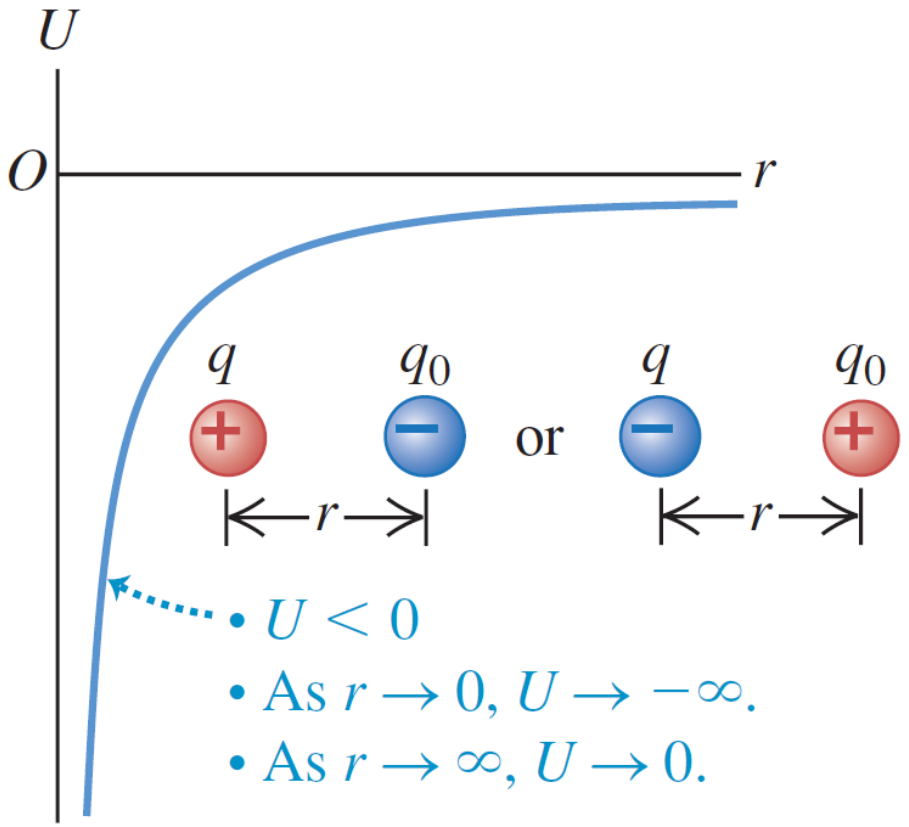
- Figure in the next slide is the graphical illustration of how U varies with these interactions.

23.1 Electric Potential Energy

(a) q and q_0 have the same sign.



(b) q and q_0 have opposite signs.



Example 23.1 Conservation of energy with electric forces

A positron (the antiparticle of the electron) has mass $9.11 \times 10^{-31}\ kg$ and charge $+e = +1.6 \times 10^{-19}\ C$. Suppose a positron moves in the vicinity of an alpha particle, which has charge $+2e = 3.2 \times 10^{-19}\ C$. The alpha particle is over 7000 times as massive as the positron, so we assume that it is at rest in some inertial frame of reference. When the positron is $1.00 \times 10^{-10}\ m$ from the alpha particle, it is moving directly away from the alpha particle at a speed of $3.00 \times 10^{6}\ m/s$.

Example 23.1 Conservation of energy with electric forces

a) What is the positron's speed when the two particles are $2.00 \times 10^{-10}\ m$ apart?

b) What is the positron's speed when it is very far away from the alpha particle?

c) How would the situation change if the moving particle were an electron (same mass as the positron but opposite charge)?

Identify

The electric force between the positron and alpha particle is conservative, so mechanical energy (kinetic plus potential) is conserved.

Set Up

The kinetic and potential energies at any two points a and b are related by $Eq.(23.3), K_a + U_a = K_b + U_b$, and the potential energy at any distance r is given by $Eq.(23.9)$.

We are given complete information about the system at a point a where the two charges are $1.00 \times 10^{-10}\, m$ apart. We use $Eq.(23.3)$ and (23.9) to find the speed at two different values of r in parts (a) and (b), and for the case where the charge $+e$ is replaced by $-e$ in part (c).

Execute

a) In this part, $r_b = 2.00 \times 10^{-10}\, m$ and we want to find the final speed v_b of the positron. This appears in the expression for the final kinetic energy, $K_b = \frac{1}{2} m {v_b}^2$; solving the energy-conservation equation for K_b, we have

$$K_b = K_a + U_a - U_b$$

The values of the energies on the right-hand side of this expression are

Execute

$$K_a = \frac{1}{2}mv_a{}^2$$
$$= \frac{1}{2}\left(9.11 \times 10^{31}\ kg\right)(3.00 \times 10^6\ m/s)^2$$
$$= 4.10 \times 10^{-18}\ J$$

$$U_a = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r_a}$$
$$= \left(9.0 \times 10^9\ N \cdot m^2/C^2\right)\frac{(3.20 \times 10^{-9}\ C)(1.60 \times 10^9\ C)}{1.00 \times 10^{-10}\ m}$$
$$= 4.61 \times 10^{-18}\ J$$

Execute

$$U_b = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r_b}$$
$$= \left(9.0 \times 10^9\ N \cdot m^2/C^2\right)\frac{(3.20 \times 10^{-9}\ C)(1.60 \times 10^9\ C)}{2.00 \times 10^{-10}\ m}$$
$$= 2.30 \times 10^{-18}\ J$$

Hence the final kinetic energy is

$$K_b = \frac{1}{2}m{v_b}^2$$
$$= 4.10 \times 10^{-18}\ J + 4.61 \times 10^{-18}\ J - 2.30 \times 10^{-18}\ J$$
$$= 6.41 \times 10^{-18}\ J$$

Execute

and the final speed of the positron is

$$v_b = \sqrt{\frac{2K_b}{m}}$$

$$= \sqrt{\frac{2(6.41 \times 10^{-18}\ J\ )}{9.11 \times 10^{31}\ kg}}$$

$$= 3.8 \times 10^6\ m/s$$

The force is repulsive, so the positron speeds up as it moves away from the stationary alpha particle.

Execute

b) When the final positions of the positron and alpha particle are very far apart, the separation r_b approaches infinity and the final potential energy U_b approaches zero. Then the final kinetic energy of the positron is

$$K_b = K_a + U_a - U_b$$
$$= 4.10 \times 10^{-18}\,J + 4.61 \times 10^{-18}\,J - 0$$
$$= 8.71 \times 10^{-18}\,J$$

Execute

And its final speed is

$$v_b = \sqrt{\frac{2K_b}{m}}$$

$$= \sqrt{\frac{2(8.71 \times 10^{-18}\ J\ )}{9.11 \times 10^{31}\ kg}}$$

$$= 4.4 \times 10^{6}\ m/s$$

Execute

Comparing to part (a), we see that as the positron moves from $r = 2.00 \times 10^{-10}\, m$ to infinity, the additional work done on it by the electric field of the alpha particle increases the speed by only about 16%.

This is because the electric force decreases rapidly with distance.

Execute

c) If the moving charge is negative, the force on it is attractive rather than repulsive, and we expect it to slow down rather than speed up.

The only difference in the above calculations is that both potential-energy quantities are negative.

From part (a), at a distance $r_b = 2.00 \times 10^{-10}\ m$ we have

Execute

$$K_b = K_a + U_a - U_b$$
$$= 4.10 \times 10^{-18}\ J + \left(-4.61 \times 10^{-18}\ J\right) - \left(-2.30 \times 10^{-18}\ J\right)$$
$$= 1.79 \times 10^{-18}\ J$$

$$v_b = \sqrt{\frac{2K_b}{m}} = 2.0 \times 10^6\ m/s$$

Execute

From part (b), at $r_b = \infty$, the kinetic energy of the electron would seem to be

$$\begin{aligned} K_b &= K_a + U_a - U_b \\ &= 4.10 \times 10^{-18}\, J + (-4.61 \times 10^{-18}\, J) - 0 \\ &= -5.1 \times 10^{-19}\, J \end{aligned}$$

But kinetic energies can never be negative!

This result means that the electron can never reach $r_b = \infty$;

Execute

The attractive force brings the electron to a halt at a finite distance from the alpha particle.

You can solve for the distance r_b at which the electron comes momentarily to rest by setting K_b equal to zero in the equation for conservation of mechanical energy (see Exercise 23.8).

Evaluate

It's useful to compare our calculations with Fig. 23.6.

In parts (a) and (b), the charges have the same sign; since $r_b > r_a$, the potential energy U_b is less than U_a.

In part (c), the charges have opposite signs; since $r_b > r_a$, the potential energy U_b is greater (that is, less negative) than U_a.

23.1 Electric Potential Energy

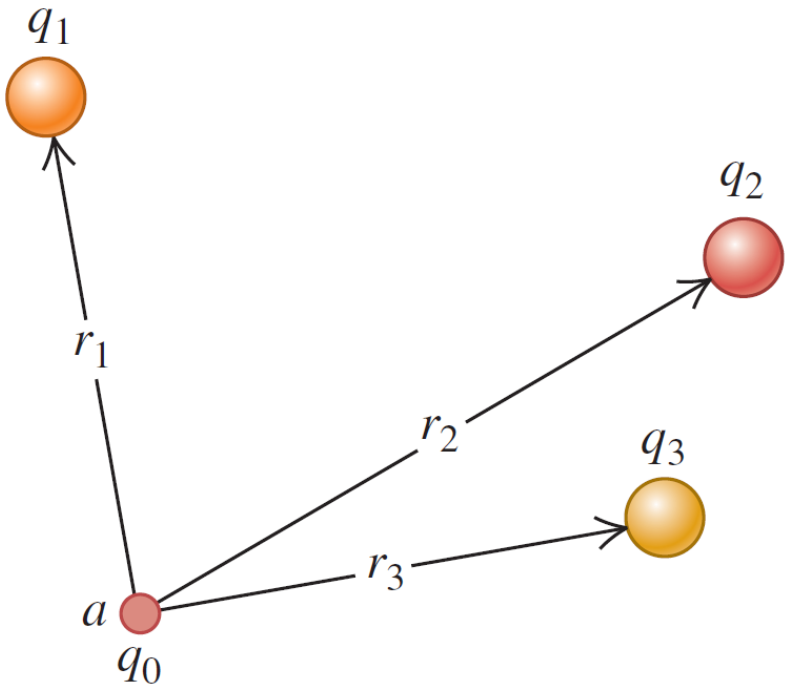
- The total electric field at each point is the vector sum of the fields due to the individual charges,
- And the total work done on the charge q_0 during any displacement is the sum of the contributions from the individual charges.

$$U = \frac{q_0}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \cdots\right) = \frac{q_0}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i} \qquad (23.10)$$

(point charge q_0 and collection of charge q_i)

23.1 Electric Potential Energy

- For every electric field due to a static charge distribution, the force exerted by that field is conservative.



Example 23.2 A system of point charges

Two point charges are located on the x-axis, $q_1 = -e$ at $x = 0$ and $q_2 = +e$ at $x = a$.

a) Find the work that must be done by an external force to bring a third point charge $q_3 = +e$ from infinity to $x = 2a$.

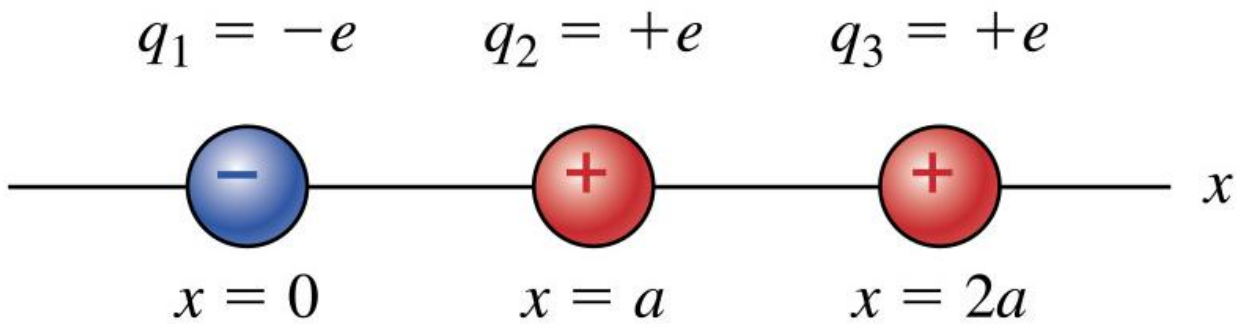
b) Find the total potential energy of the system of three charges.

Identify and Set Up

Figure below shows the final arrangement of the three charges.

To find the work required to bring q_3 in from infinity, we can find the potential energy associated with q_3 in the presence of q_1 and q_2.

We then can find the total potential energy of the system.



Execute

a) The work that must be done on q_3 by an external force $\vec{\boldsymbol{F}}_{ext}$ is equal to the difference between two quantities: the potential energy U associated with q_3 when it is at $x = 2a$ and the potential energy when it is infinitely far away. The second of these is zero, so the work that must be done is equal to U.

The distances between the charges are $r_{13} = 2a$ and $r_{23} = a$, so from Eq. (23.10),

Execute

$$W = U = \frac{q_3}{4\pi\varepsilon_0}\left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}}\right) = \frac{+e}{4\pi\epsilon_0}\left(\frac{-e}{2a} + \frac{+e}{a}\right) = \frac{+e^2}{8\pi\varepsilon_0 a}$$

If q_3 is brought in from infinity along the $+x$-axis, it is attracted by q_1 but is repelled more strongly by q_2; hence positive work done must be done to push q_3 to the position at $x = 2a$.

Execute

b) The total potential energy of the three charge system is

$$U = \frac{1}{4\pi\varepsilon_0}\sum_{i<j}\frac{q_i q_j}{r_{ij}}$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right)$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{(-e)(e)}{a} + \frac{(-e)(e)}{2a} + \frac{(e)(e)}{a}\right)$$

$$= \frac{-e^2}{8\pi\varepsilon_0 a}$$

Evaluate

Since our result in part (b) is negative, the system has lower potential energy than it would if the three charges were infinitely far apart.

An external force would have to do negative work to bring the three charges from infinity to assemble this entire arrangement and would have to do positive work to move the three charges back to infinity.

Chapter 23.2

Electric Potential

23.2 Electric Potential

- Generally, potential is the potential energy per unit charge.
- SI units: volt ($1\ V$).
- V_{ab}, the potential of a with respect to b, is the work done by the electric force when a UNIT charge moves from a to b.
- In other words, it is also the work that must be done against the electric force to move a UNIT charge slowly from b to a.

23.2 Electric Potential

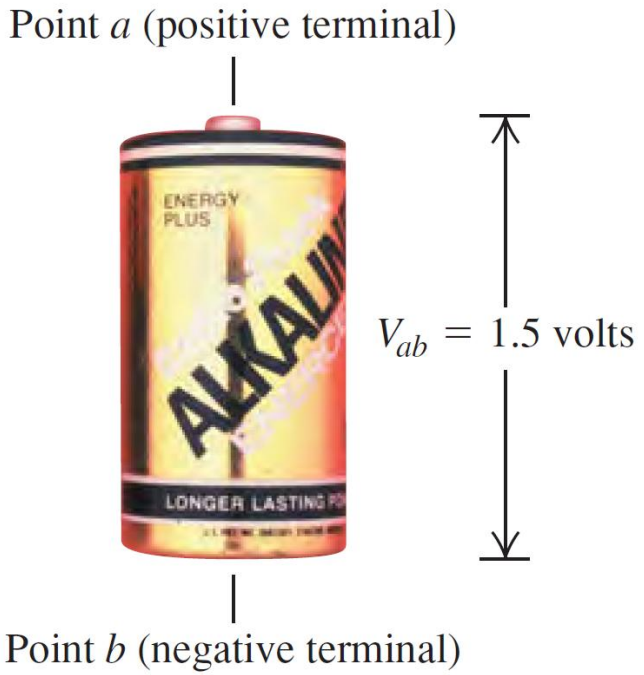
- In electric circuits, the potential difference between two points is often called voltage.

$$V = \frac{U}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \qquad (23.14)$$

(potential due to a point charge)

$$V = \frac{U}{q_0} = \frac{1}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i} \qquad (23.15)$$

(potential due to a collection of point charges)



23.2 Electric Potential

When we have a continuous distribution of charge along a line, over a surface, or through a volume

$$V = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r} \qquad (23.16)$$

(potential due to a continuous distribution of charge)

23.2 Electric Potential

The work done by the electric force as the test charge moves from a to b is given by

$$W_{a\to b} = \int_a^b \vec{\boldsymbol{F}} \cdot d\vec{\boldsymbol{l}} = \int_a^b q_0 \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}}$$

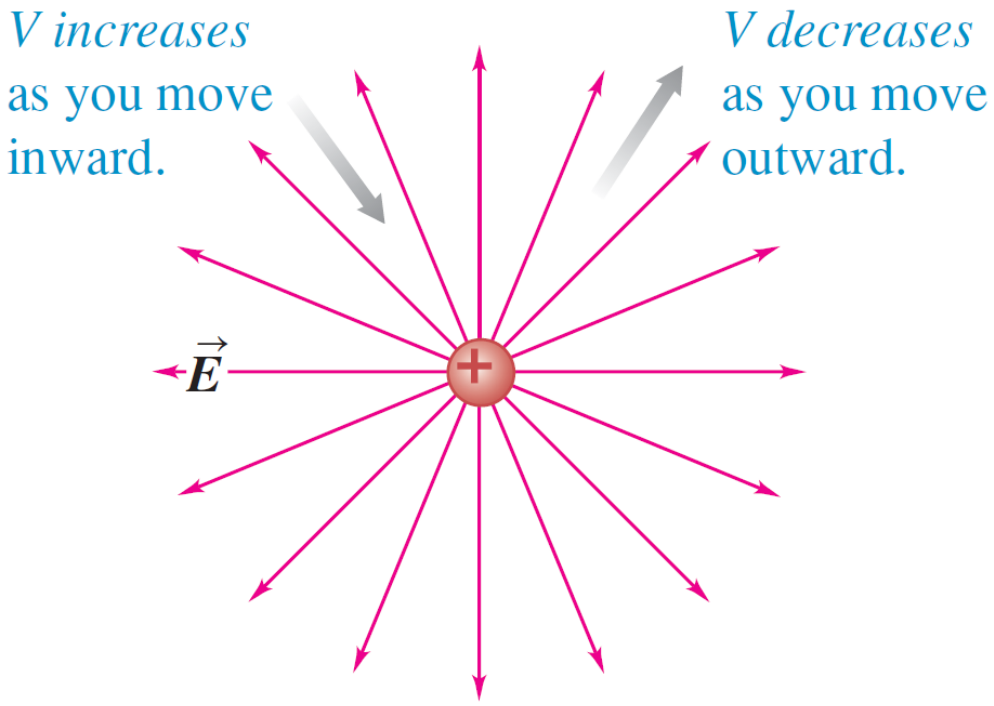
If we divide this by q_0 and we can find

$$V_a - V_b = \int_a^b \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = \int_a^b E\cos\varphi\, dl \qquad (23.17)$$

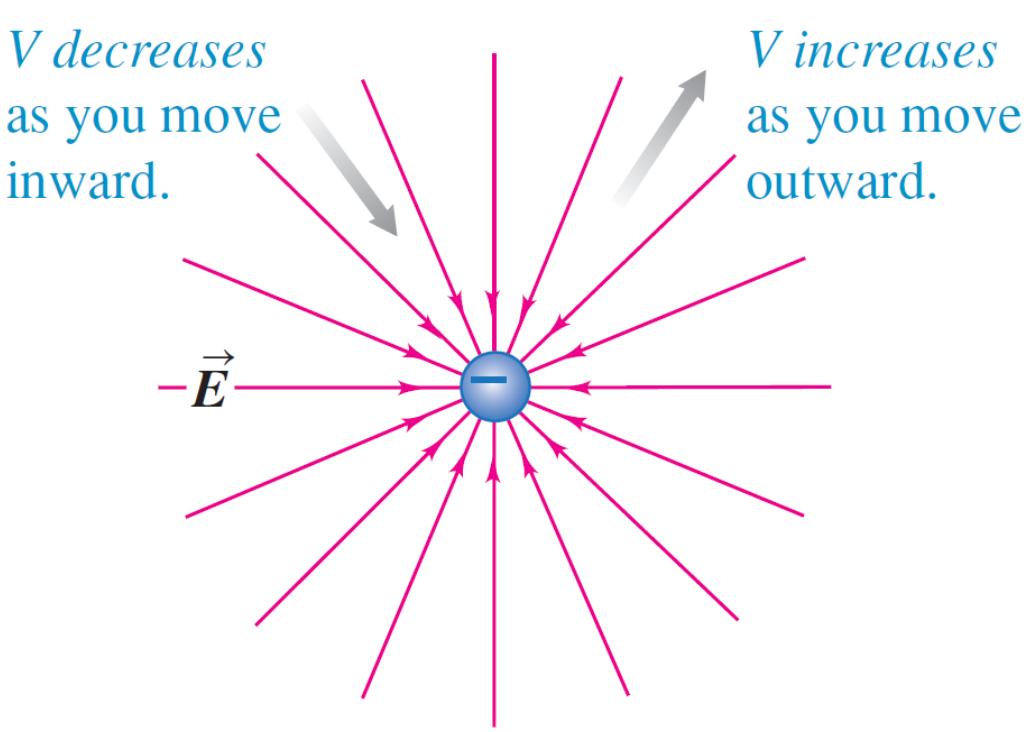
(potential difference as an integral $\vec{\boldsymbol{E}}$)

23.2 Electric Potential

(a) A positive point charge



(b) A negative point charge



Electron Volts

- The magnitude e of the electron charge can be used to define a unit of energy that is useful in many calculations with atomic and nuclear systems.

- When a particle with charge q moves from a point where the potential is V_b to a point where it is V_a, the change in the potential energy U is

$$U_a - U_b = q(V_a - V_b) = qV_{ab}$$

Electron Volts

- If charge q equals the magnitude e of the electron charge, $1.602 \times 10^{-19}\,C$, and the potential difference is $V_{ab} = 1\,V$, the change in energy is

$$U_a - U_b = (1.602 \times 10^{-19}\,C)(1\,V) = 1.602 \times 10^{-19}\,J$$

- This quantity of energy is defined to be 1 **electron volt** (1 ev):

$$1\,ev = 1.602 \times 10^{-19}\,J$$

- The multiples meV, keV, MeV, GeV, and TeV are often used.

Example 23.3 Electric force and electric potential

A proton (charge $+e = 1.602 \times 10^{-19}\ C$) moves in a straight line from point a to point b inside a linear accelerator, a total distance $d = 0.50\ m$. The electric field is uniform along this line, with magnitude $E = 1.5 \times 10^{7}\ V/m = 1.5 \times 10^{7}\ N/V$ in the direction from a to b.

Determine

a) the force on the proton;
b) the work done on it by the field;
c) the potential difference $V_a - V_b$.

Identify and Set Up

We are given the electric field, so it is straightforward to find the electric force on the proton.

Calculating the work done on the proton by this force is also straightforward because $\vec{E}$ is uniform, which means that the force is constant.

Once the work is known, we find the potential difference using Eq. (23.13).

Execute

a) The force is in the same direction as the electric field, and its magnitude is

$$\begin{aligned} F &= qE \\ &= \left(1.602 \times 10^{-19}\ C\right)\left(1.5 \times 10^{7}\ N/C\right) \\ &= 2.4 \times 10^{-12}\ N \end{aligned}$$

Execute

b) The force is constant and in the same direction as the displacement, so the work done is

$$
\begin{aligned}
W_{a\to b} &= Fd \\
&= \left(2.4\times10^{-12}\ N\right)(0.50\ m) \\
&= 1.2\times10^{-12}\ J \\
&= \left(1.2\times10^{-12}\ J\right)\frac{1\ eV}{1.602\times10^{-19}\ J} \\
&= 7.5\times10^{6}\ ev \\
&= 7.5\ Mev
\end{aligned}
$$

Execute

c) From Eq. (23.13) the potential difference is the work per unit charge, which is

$$\begin{aligned} V_a - V_b &= \frac{W_{a\to b}}{q} \\ &= \frac{1.2\times10^{-12}\ J}{1.602\times10^{-19}\ C} \\ &= 7.5\times10^{6}\ J/C \\ &= 7.5\times10^{6}\ V \\ &= 7.5\ MV \end{aligned}$$

Execute

We can get this same result even more easily by remembering that 1 electron volt equals 1 volt multiplied by the charge e.

Since the work done is $7.5 \times 10^6\ eV$ and the charge is $1e$, the potential difference is $(7.5 \times 10^6\ eV)/e = 7.5 \times 10^6\ V$.

Evaluate

We can check our result in part (c) by using Eq. (23.17) or (23.18) to calculate an integral of the electric field.

The angle ϕ between the constant field $\vec{E}$ and the displacement is zero, so Eq. (23.17) becomes

$$V_a - V_b = \int_a^b E\cos\phi\, dl = \int_a^b E dl = E\int_a^b dl$$

Evaluate

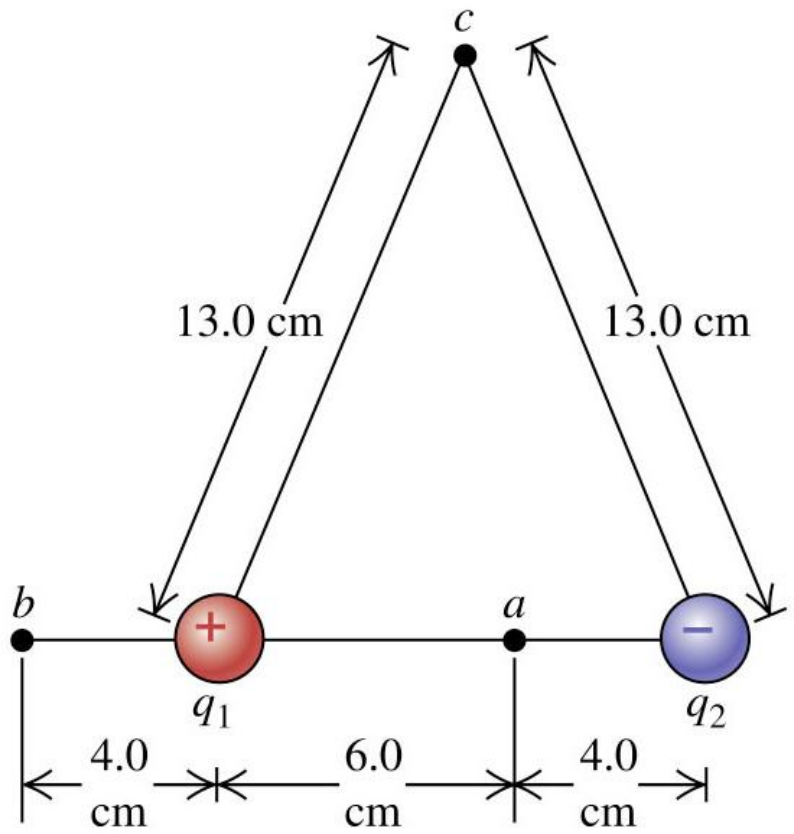
The integral of dl from a to b is just the distance d, so we again find

$$V_a - V_b = Ed = (1.5 \times 10^7\ V/m)(0.50\ m) = 7.5 \times 10^6\ V$$

Example 23.4 Potential due to two point charges

An electric dipole consists of two point charges, $q_1 = +12\,nC$ and $q_2 = -12\,nC$, placed $10\,cm$ apart (Fig. 23.13).

Compute the potentials at points a, b, and c.



Identify and Set Up

This is the same arrangement of charges as in Example 21.9 (Section 21.5). In that example we calculated electric field at each point by doing a vector sum.

Here we calculate the electric potential at each point by doing the algebraic sum in Eq. (23.15):

$$V = \frac{1}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i}$$

Execute

At point a the potential due to the positive charge q_1 is

$$\frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1} = \left(9.0 \times 10^9\ N \cdot m^2/C^2\right)\frac{12 \times 10^{-9}\ C}{0.060\ m}$$

$$= 1800\ N \cdot \frac{m}{C}$$

$$= 1800\,\frac{J}{C}$$

$$= 1800\ V$$

Execute

and the potential due to the negative charge q_2 is

$$\frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2} = \left(9.0\times10^9\ N\cdot m^2/C^2\right)\frac{-12\times10^{-9}\ C}{0.040\ m}$$

$$= -2700\ N\cdot\frac{m}{C}$$

$$= -2700\frac{J}{C}$$

$$= -2700\ V$$

Execute

The potential V_a at point a is the sum of these:

$$V_a = 1800\,V + (-2700\,V) = -900\,V$$

By similar calculations you can show that at point b the potential due to the positive charge is $+2700\,V$, the potential due to that negative charge is $-770\,V$, and

$$V_b = 2700\,V + (-770\,V) = 1930\,V$$

Execute

At point c the potential due to the positive charge is

$$\frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1} = \left(9.0 \times 10^9\ N \cdot m^2/C^2\right)\frac{12 \times 10^{-9}\ C}{0.060\ m} = 830\ V$$

The potential due to the negative charge is $-830\ V$, and the total potential is zero:

$$V_c = 830\ V + (-830\ V) = 0$$

Execute

The potential is also equal to zero at infinity (infinitely far from both charges).

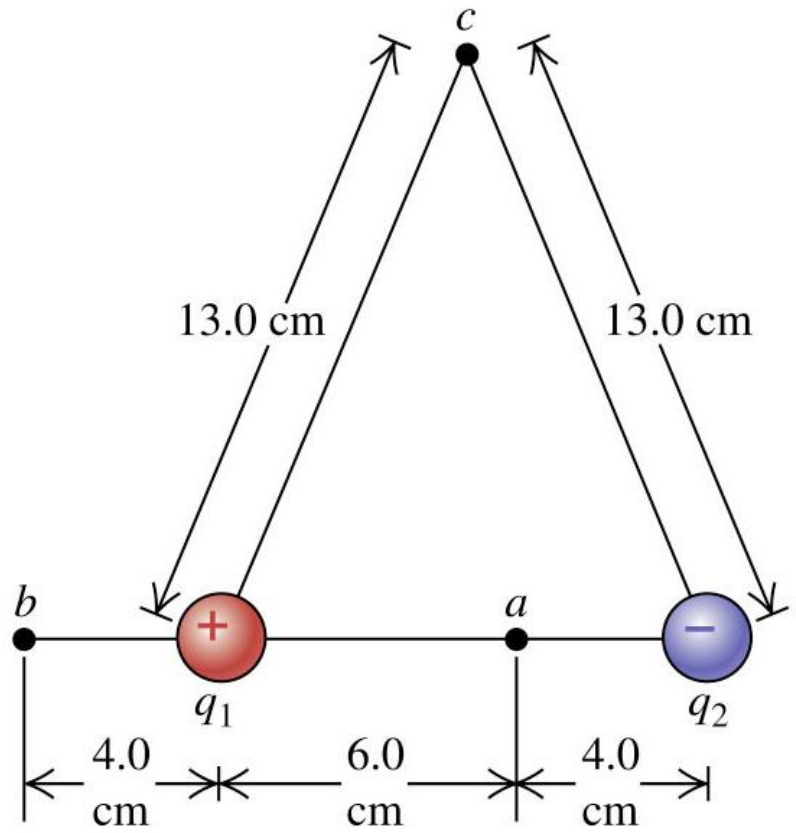
Evaluate

Comparing this example with Example 21.9 shows that it's much easier to calculate electric potential (a scalar) than electric field (a vector).

We'll take advantage of this simplification whenever possible.

Example 23.5 Potential and potential energy

Compute the potential energy associated with a point charge of $+4.0\,nC$ if it is placed at points a, b, and c in the figure on the right side.



Identify and Set Up

For any point charge q, the associated potential energy is $U = qV$. We use the values of V from Example 23.4.

Execute

At point a,

$$U_a = qV_a = \left(4.0 \times 10^{-9}\ C\right)(-900\ J/C) = -3.6 \times 10^{-6}\ J$$

At point b,

$$U_b = qV_b = \left(4.0 \times 10^{-9}\ C\right)(1930\ J/C) = 7.7 \times 10^{-6}\ J$$

At point c,

$$U_c = qV_c = 0$$

All of these values correspond to U and V being zero at infinity.

Evaluate

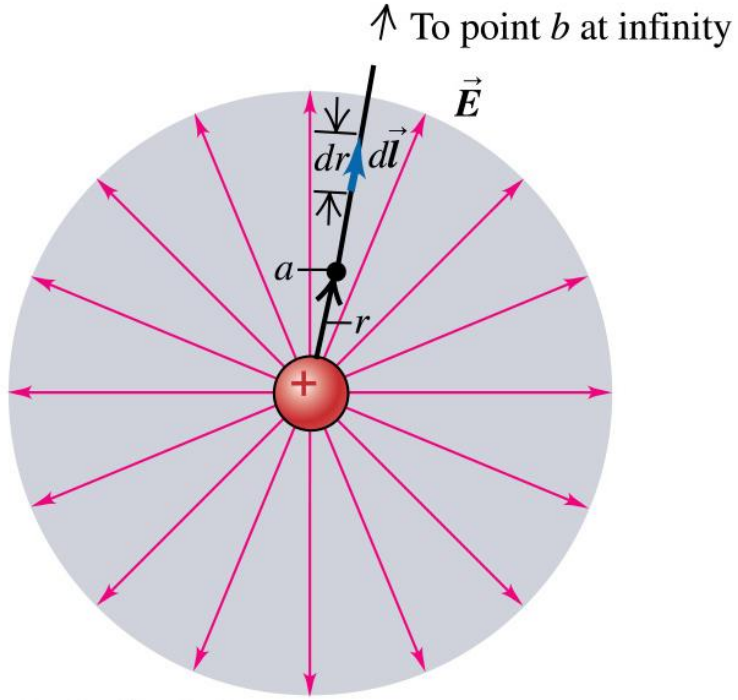
- Note that no net work is done on the $4.0 - nC$ charge if it moves from point c to infinity by any path.
- In particular, let the path be along the perpendicular bisector of the line joining the other two charges q_1 and q_2 in Fig. 23.13.
- As shown in Example 21.9 (section 21.5), at points on the bisector the direction of $\vec{E}$ is perpendicular to the bisector.
- Hence the force on the $4.0\,nC$ charge is perpendicular to the path, and no work is done in any displacement along it.

Evaluate

- Hence the force on the $4.0\,nC$ charge is perpendicular to the path, and no work is done in any displacement along it.

Example 23.6 Finding potential by integration

By integrating the electric field as in Eq. (23.17), find the potential at a distance r from a point charge q.



Identify and Set Up

To find the potential V at a distance r from the point charge, we let point a in Eq. (23.17) be at distance r and let point b be at infinity (Fig. 23.14).

As usual, we choose the potential to be zero at an infinite distance from the charge.

Execute

To carry out the integral, we can choose any path we like between points a and b.

The most convenient path is a straight radial line as shown in Fig. 23.14, so that $d\vec{l}$ is in the radial direction and has magnitude dr.

If q is positive, $\vec{E}$ and $d\vec{l}$ are always parallel, so $\phi = 0$ and Eq. (23.17) becomes

Execute

$$V - 0 = V = \int_r^\infty \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = \int_r^\infty E dr = \int_r^\infty \frac{q}{4\pi\varepsilon_0 r^2} dr$$

$$= -\frac{q}{4\pi\varepsilon_0 r}\Big|_r^\infty = 0 - \left(-\frac{q}{4\pi\varepsilon_0 r}\right) = \frac{q}{4\pi\varepsilon_0 r}$$

This agrees with Eq. (23.14). If q is negative, $\vec{\boldsymbol{E}}$ is radially inward while $d\vec{\boldsymbol{l}}$ is still radially outward, so $\phi = 180°$.

Execute

Since $\cos 180° = -1$, this adds a minus sign to the above result.

However, the field magnitude E is always positive, and since q is negative, we must write $E = |q|/4\pi\varepsilon_0 r$, giving another minus sign.

The two minus signs cancel, and the above result for V is valid for point charges of either sign.

Evaluate

We can get the same result by using Eq. (23.7) for the electric field, which is valid for either sign of q, and writing $d\vec{\boldsymbol{l}} = \hat{\boldsymbol{r}}dr$:

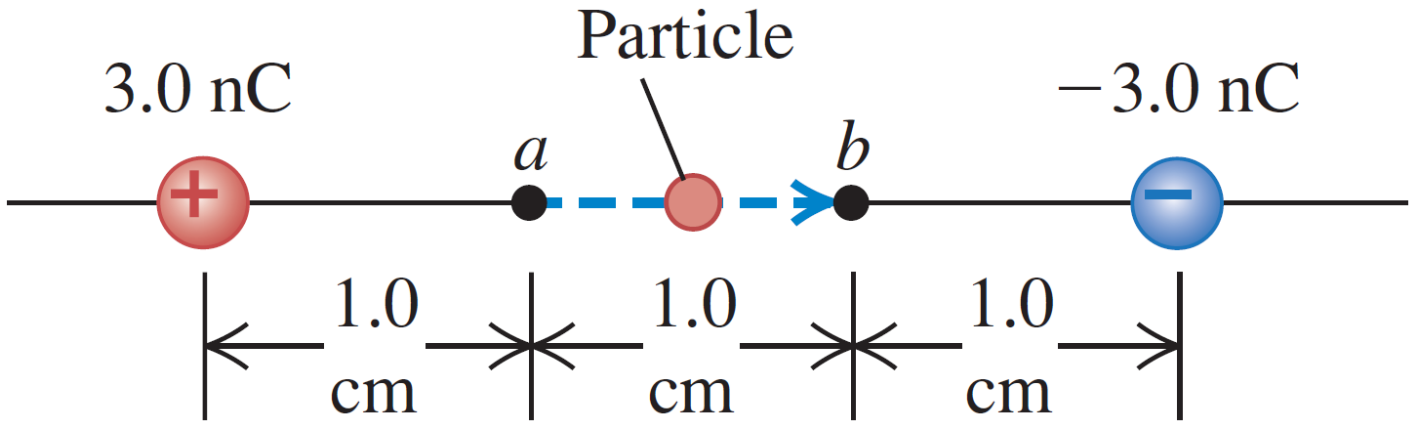
$$V - 0 = V = \int_r^\infty \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}}$$

$$= \int_r^\infty \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{r}}dr = \int_r^\infty \frac{q}{4\pi\varepsilon_0 r^2}dr$$

Example 23.7 Moving through a potential difference

In the figure below a dust particle with mass $m = 5.0 \times 10^{-9}\, kg = 5.0\, \mu g$ and charge $q_0 = 2.0\, nC$ starts from rest at point a and moves in a straight line to point b.

What is its speed v at point b?



Identify and Set Up

This problem would be difficult to solve without using energy techniques, since the force that acts on the particle varies in magnitude as the particle moves from a to b.

Only the conservative electric force acts on the particle, so mechanical energy is conserved:

$$K_a + U_a = K_b + U_b$$

Execute

For this situation, $K_a = 0$ and $K_b = \frac{1}{2}mv^2$.

We get the potential energies (U) from the potentials (V) using Eq. (23.12): $U_a = q_0 V_a$ and $U_b = q_0 V_b$. Substituting these into the energy-conservation equation and solving for v, we find

$$0 + q_0 V_a = \frac{1}{2}mv^2 + q_0 V_b$$

$$v = \sqrt{\frac{2q_0(V_a - V_b)}{m}}$$

Execute

We calculate the potentials using Eq. (23.15), just as we did in Example 23.4:

$$V_a = \left(9.0 \times 10^9\ N \cdot m^2/C^2\right) \times \left(\frac{3.0 \times 10^{-9}\ C}{0.010\ m} + \frac{-3.0 \times 10^{-9}\ C}{0.020\ m}\right)$$
$$= 1350\ V$$

$$V_b = \left(9.0 \times 10^9\ N \cdot m^2/C^2\right) \times \left(\frac{3.0 \times 10^{-9}\ C}{0.020\ m} + \frac{-3.0 \times 10^{-9}\ C}{0.010\ m}\right)$$
$$= -1350\ V$$

Execute

$$V_a - V_b = (1350\,V) - (-1350\,V) = 2700\,V$$

$$v = \sqrt{\frac{2(2.0 \times 10^{-9}\,C)(2700\,V)}{5.0 \times 10^{-9}\,kg}} = 46\,m/s$$

Evaluate

- Our result makes sense: the positive test charge gains speed as it moves away from the positive charge and toward the negative charge.
- To check unit consistency in the final line of the calculation, note that $1V = 1J/C$, so the numerator under the radical has units of J or $kg \cdot m^2/s^2$.

Evaluate

- We can use exactly this same method to find the speed of an electron accelerated across a potential difference of $500\,V$ in an oscilloscope tube or $20\,kV$ in a TV picture tube.
- The end-of-chapter problems include several examples of such calculations.

23.3 Calculating Electric Potential

- When calculating the potential due to a charge distribution, we usually follow one of two routes.

  - If we know the charge distribution, we can use $V = \frac{U}{q_0} = \frac{1}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i}$ or $V = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r}$.

  - Or if we know how the electric field depends on position, we can use $V_a - V_b = \int_a^b \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = \int_a^b E\cos\varphi\, dl$, defining the potential to be zero at some convenient place.

23.3 Calculating Electric Potential

- Some problems require a combination of these approaches.
- You'll see how much easier it is to calculate scalar electric potentials than vector electric fields. The moral is clear:
- Whenever possible, solve problems using an energy approach (using electric potential and electric potential energy) rather than a dynamics approach (using electric fields and electric forces).

Example 23.8 A charged conducting sphere

A solid conducting sphere of radius R has a total charge q. Find the potential energy everywhere, both outside and inside the sphere.

Identify and Set Up

In Example 22.5 we used Gauss's law to find the electric field at all points for this charge distribution. We can use that result to determine the potential.

Execute

the field outside the sphere is the same as if the sphere were removed and replaced by a point charge q. We take $V = 0$ at infinity, as we did for a point charge. Then the potential at a point outside the sphere at a distance r from its center is the same as that due to a point charge q at the center:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

The potential at the surface of the sphere is $V_{surface} = q/4\pi\varepsilon_0 R$.

Execute

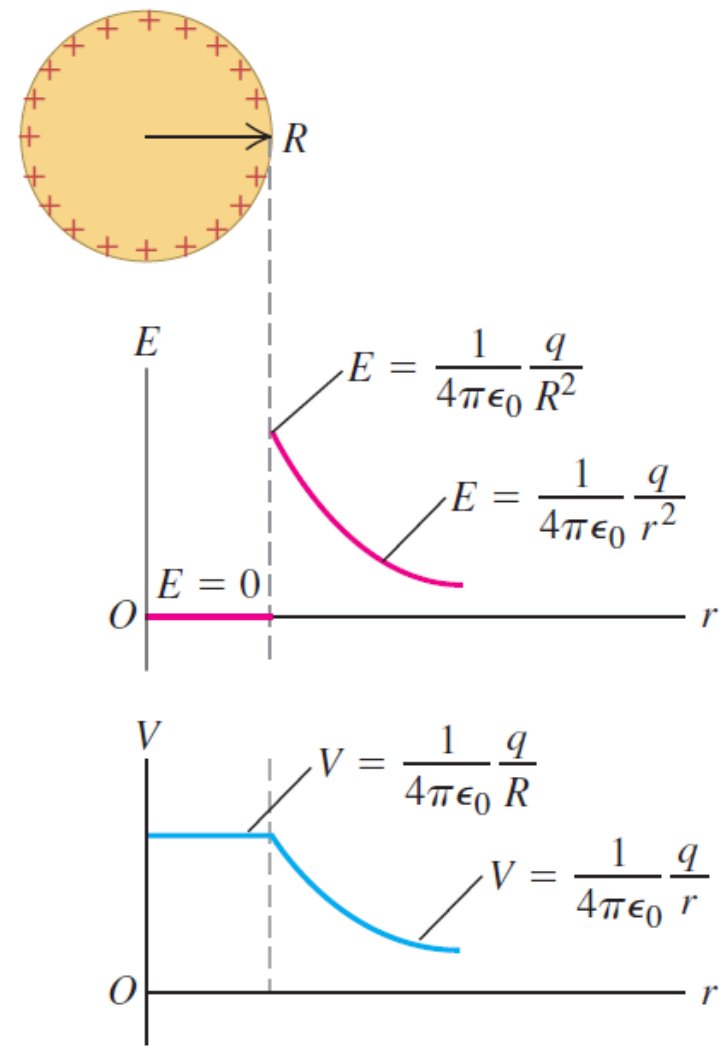
Inside the sphere, $\vec{E}$ is zero everywhere. Hence no work is done on a test charge that moves from any point to any other point inside the sphere. Thus the potential is the same at every point inside the sphere and is equal to its value $q/4\pi\varepsilon_0 R$ at the surface.

Evaluate

The figure in the next slide shows the field and potential for a positive charge q. In this case the electric field points radially away from the sphere. As you move away from the sphere, in the direction of $\vec{E}$, V decreases (as it should).

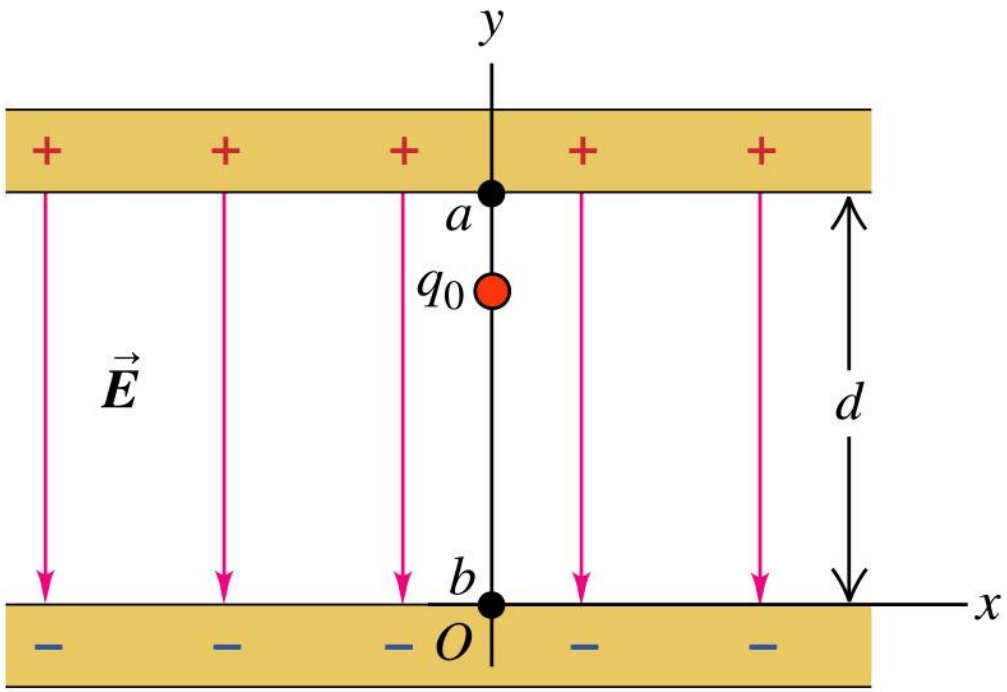
Evaluate

The figure in the next slide shows the field and potential for a positive charge q. In this case the electric field points radially away from the sphere. As you move away from the sphere, in the direction of $\vec{E}$, V decreases (as it should).


R
E
$E = \frac{1}{4\pi\epsilon_0}\frac{q}{R^2}$
$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$
$E = 0$
O
r
V
$V = \frac{1}{4\pi\epsilon_0}\frac{q}{R}$
$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$
O
r

Example 23.9 Oppositely charged parallel plates

Find the potential at any height y between the two oppositely charged parallel plates discussed in Section 23.1.



Identify and Set Up

The potential energy U for a test charge q_0 at a point a distance y above the bottom plate (Fig. 23.18) is given by Eq. (23.5), $U = q_0 E y$.

We use this expression to determine the potential V at such a point.

Execute

The potential $V(y)$ at coordinate y is the potential energy per unit charge:

$$V(y) = \frac{U(y)}{q_0} = \frac{q_0 E y}{q_0} = Ey$$

We have chosen $U(y)$, and therefore $V(y)$, to be zero at point b, where $y = 0$.

Even if we choose the potential to be different from zero at , it is still true that

Execute

$$V(y) - V_b = Ey$$

The potential decreases as we move from the upper to the lower plate in the direction of $\vec{E}$. At point a, where $y = d$ and $V(y) = V_a$,

$$V_a - V_b = Ed$$

$$\therefore E = \frac{V_a - V_b}{d} = \frac{V_{ab}}{d}$$

Execute

- Where V_{ab} is the potential of the positive plate with respect to the negative plate.
- That is, the electric field equals the potential difference between the plates divided by the distance between them.
- For a given potential difference V_{ab}, the smaller the distance d between the two plates, the greater the magnitude E of the electric field

Execute

- This relationship between E and V_{ab} holds only for the planar geometry we have described.
- It does not work for situations such as concentric cylinders or spheres in which the electric field is not uniform.

Evaluate

- Our result provides a practical way to measure the charge density on the charges on the two plates in Fig. 23.18.
- In Example 22.8 (Section 22.4), we derived the expression $E = \sigma/\varepsilon_0$ for the electric field E between two conducting plates having surface charge densities $+\sigma$ and $+\sigma$, respectively.
- Setting this expression equal to $E = V_{ab}/d$ gives

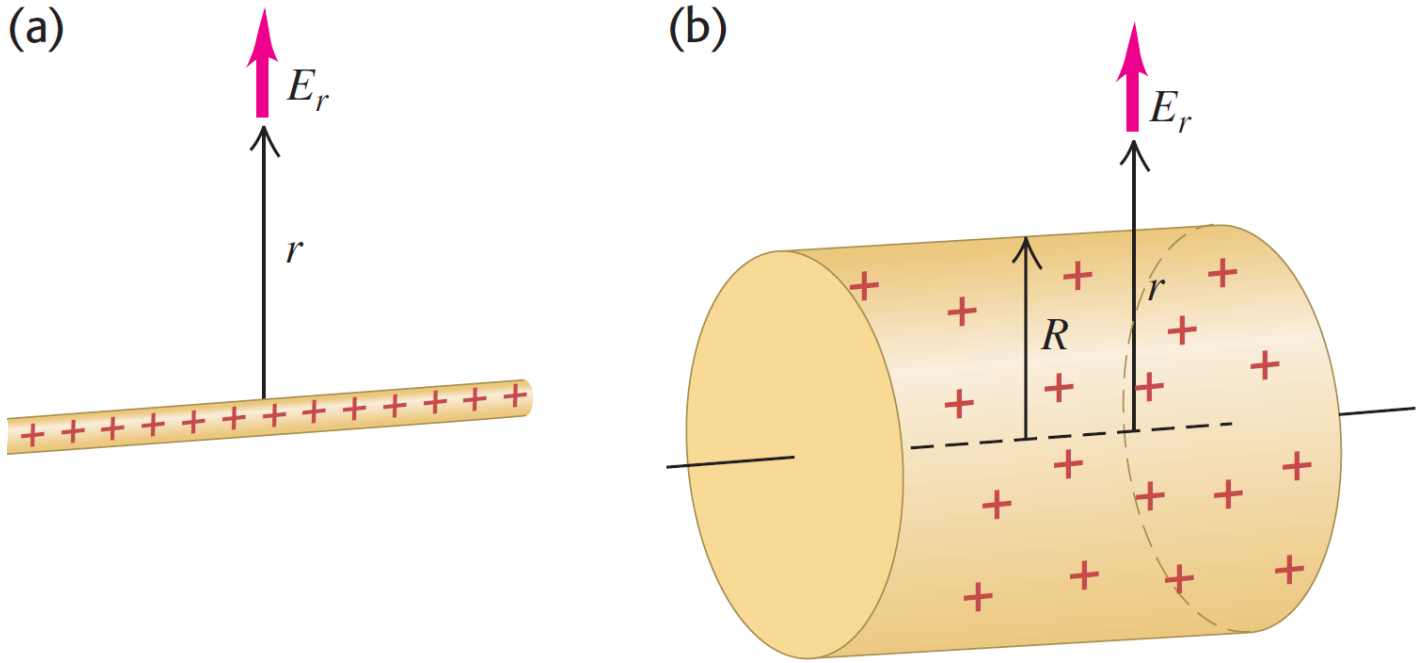
$$\sigma = \frac{\varepsilon_0 V_{ab}}{d}$$

Evaluate

- The surface charge density on the positive plate is directly proportional to the potential difference between the plates, and its value σ can be determined by measuring V_{ab}.
- This technique is useful because no instruments are available that read surface charge density directly.
- On the negative plate the surface charge density is $-\sigma$.

Example 23.10 An infinite line charge or charged conducting cylinder

Find the potential at a distance r from a very long line of charge with linear charge density (charge per unit length) λ.



Identify and Set Up

One approach to this problem is to divide the line of charge into infinitesimal elements, as we did in Example 21.11 (Section 21.5) to find the electric field produced by such a line.

We could then integrate to find the net potential V.

In this case, however, our task is greatly simplified because we already know the electric field.

Identify and Set Up

In both Example 21.11 and Example 22.6 (Section 22.4), we found that the electric field at a distance r from a straight-line charge (Fig. 23.19a) has only a radial component, given by

$$E = \frac{\lambda}{2\pi\varepsilon_0}\frac{1}{r}$$

We use this expression to find the potential by integrating $\vec{E}$ as in Eq. (23.17).

Execute

Since the field has only a radial component, the scalar product $\vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}}$ is equal to $E_r dr$.

Hence the potential of any point a with respect to any other point b, at radial distances r_a and r_b from the line of charge, is

$$V_a - V_b = \int_a^b \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = \int_a^b E_r dr = \frac{\lambda}{2\pi\varepsilon_0} ln \frac{r_b}{r_a}$$

Execute

If we take point b at infinity and set $V_b = 0$, we find that V_a is infinite:

$$V_a = \frac{\lambda}{2\pi\varepsilon_0} ln \frac{\infty}{r_a} = \infty$$

This shows that if we try to define V to be zero at infinity, then V must be infinite at any finite distance from the line charge.

Hence this is not a useful way to define V for this problem!

Execute

The difficulty, as we mentioned earlier, is that the charge distribution itself extends to infinity.

To get around this difficulty, remember that we can define V to be zero at any point we like.

We set $V_b = 0$ at point b at an arbitrary radial distance r_0. Then the potential $V = V_a$ at point a at a radial distance r is given by $V - 0 = \left(\frac{\lambda}{2\pi\varepsilon_0}\right)\ln(\frac{r_0}{r})$, or $V = \left(\frac{\lambda}{2\pi\varepsilon_0}\right)\ln(\frac{r_0}{r})$,

Evaluate

- According to our result, if λ is positive then V decreases as r increases.
- This is as it should be: V decreases as we move in the direction of $\vec{E}$.
- From Example 22.6, the expression for E_r with which we started also applies outside a long charged conducting cylinder with charge per unit length λ (Fig. 23.19b).

Evaluate

- Hence our result also gives the potential in the field for such a cylinder, but only for values of r to be the cylinder radius R, so that $V = 0$ when $r = R$, then at any point for which $r > R$,

$$V = \left(\frac{\lambda}{2\pi\varepsilon_0}\right)\ln(\frac{r_0}{r})$$

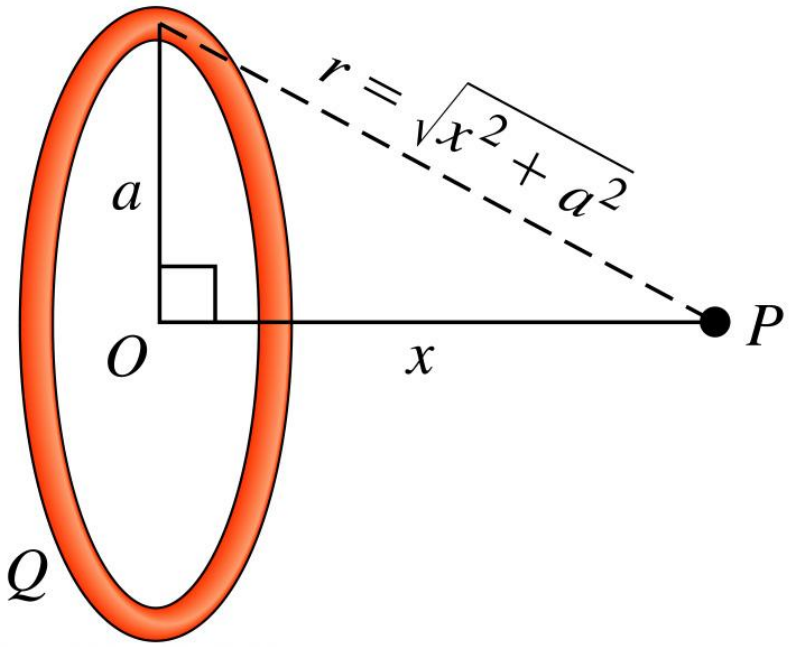
where r is the distance from the axis of the cylinder.

- Inside the cylinder, $\vec{E} = \mathbf{0}$,and V has the same value (zero) as on the cylinder's surface.

Example 23.11 A ring of charge

Electric charge is distributed uniformly around a thin ring of radius a, with total charge Q (Fig. 23.20).

Find the potential at a point P on the ring axis at a distance x from the center of the ring.



Identify and Set Up

We could solve this problem using the same approach as in the preceding example: we found the electric field at all points along the x-axis in Example 21.10 (Section 21.5), so we could integrate $\vec{E}$ as in Eq. (23.17) to find at V along this axis.

But in this case, however, it is far easier to find V by integrating over the charge distribution as in Eq. (23.16), because all parts of the ring (that is, all elements of the charge distribution) are the same distance r from point P.

Execute

We divide the ring into infinitesimal elements of charge dq. Figure above shows that the distance from each such element to the point P is $r = \sqrt{x^2 + a^2}$. Hence we can take the factor $1/r$ outside the integral in Eq. (23.16), and

$$V = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r} = \frac{1}{4\pi\varepsilon_0}\frac{1}{r}\int dq$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{1}{\sqrt{x^2 + a^2}}\int dq = \frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{x^2 + a^2}}$$

Execute

Potential is a scalar quantity; there is no need to consider components of vectors in this calculation, as we had to do when we found the electric field at P.

So the potential calculation is a lot simpler than the field calculation.

Evaluate

- Note that when x is much larger than a, the above expression for V becomes approximately equal to $V = Q/4\pi\varepsilon_0 x$.
- This corresponds to the potential of a point charge Q at a distance x.
- When we are very far away from a charged ring, it looks like a point charge. (We drew a similar conclusion about the electric field of a ring in Example 21.10.)

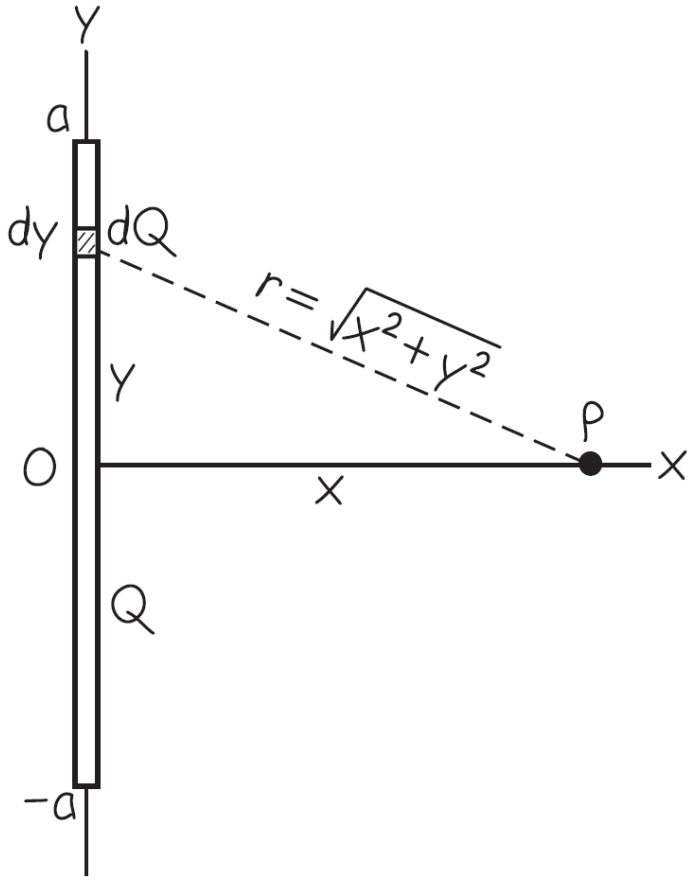
Evaluate

These results for V can also be found by integrating the expression for E_x, found in Example 21.10 (see Problem 23.67).

Example 23.12 A line of charge

Electric charge Q is distributed uniformly along a line or thin rod of length $2\,a$.

Find the potential at a point P along the perpendicular bisector of the rod at a distance x from its center.



Identify and Set Up

This is the same situation as in Example 21.11 (Section 21.5), where we found an expression for the electric field $\vec{E}$ at an arbitrary point on the x-axis. We could integrate $\vec{E}$ using Eq. (23.17) to find V.

Instead, we'll integrate over the charge distribution using Eq. (23.16) to get a bit more experience with this approach.

Execute

The element of charge dQ corresponding to an element of length dy on the rod is given by $dQ = (Q/2a)dy$ (Figure above).

The distance from dQ to P is $\sqrt{x^2 + y^2}$, and the contribution dV that it makes to the potential at P is

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\frac{dy}{\sqrt{x^2 + y^2}}$$

Execute

To get the potential at P due to the entire rod, we integrate dV over the length of the rod from $y = -a$ to $y = a$:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\int \frac{dy}{\sqrt{x^2 + y^2}}$$

You can look up the integral in a table. The final result is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\ln\left(\frac{\sqrt{a^2 + x^2} + a}{\sqrt{a^2 + x^2} - a}\right)$$

Evaluate

- We can check our result by letting x approach infinity. In this limit the point P is infinitely far from all of the charge, so we expect V to approach zero; we invite you to verify that it does so.

- As in Example 23.11, this problem is simpler than finding at point P because potential is a scalar quantity and no vector calculations are involved.

23.4 Equipotential Surfaces

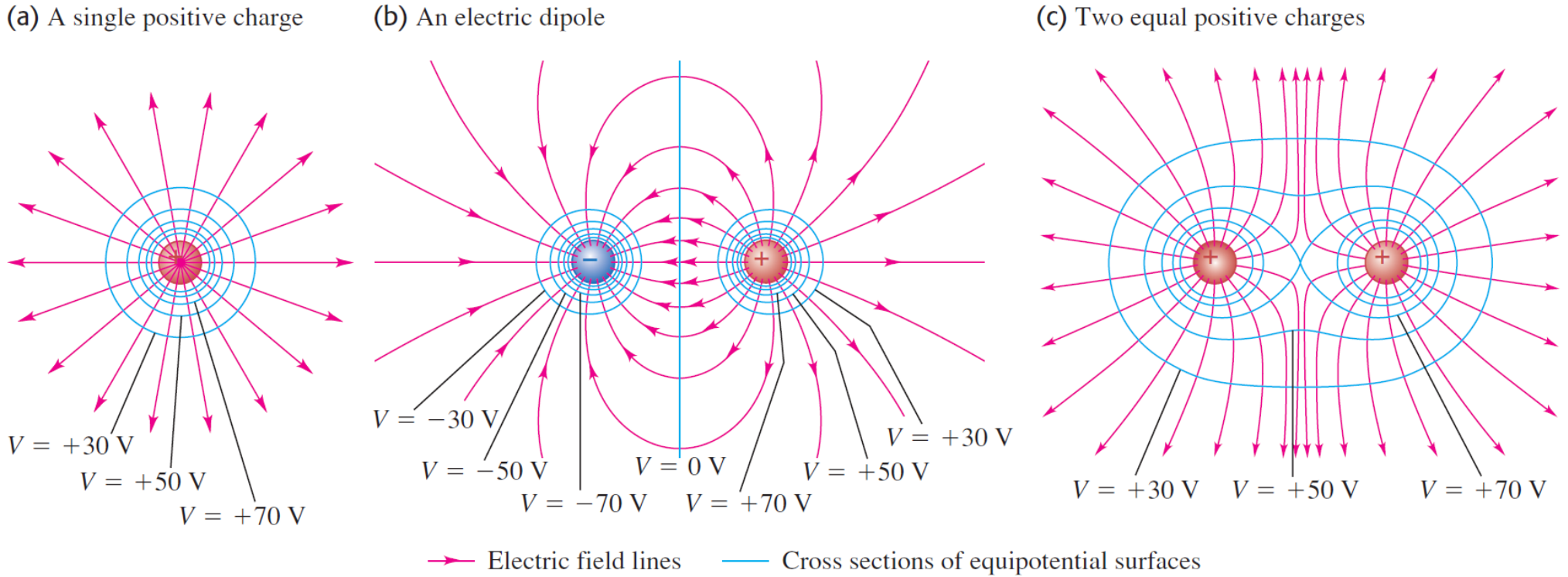
- An equipotential surface is a three-dimensional surface on which the electric potential V is the same at every point.
- Field lines and equipotential surfaces are always mutually perpendicular.
- When all charges are at rest, the surface of a conductor is always an equipotential surface, and the electric field just outside it must be perpendicular to the surface at every point.

23.4 Equipotential Surfaces


Oso
Canyon
Lower Oso Campground
Los
Prietos
Boys Camp
Lewis
Canyon
PARADISE RD
1000
1200
1400
1600
1800
1200
1400
1600
1800
1400
1600
1800
1400
1600
1800

23.4 Equipotential Surfaces

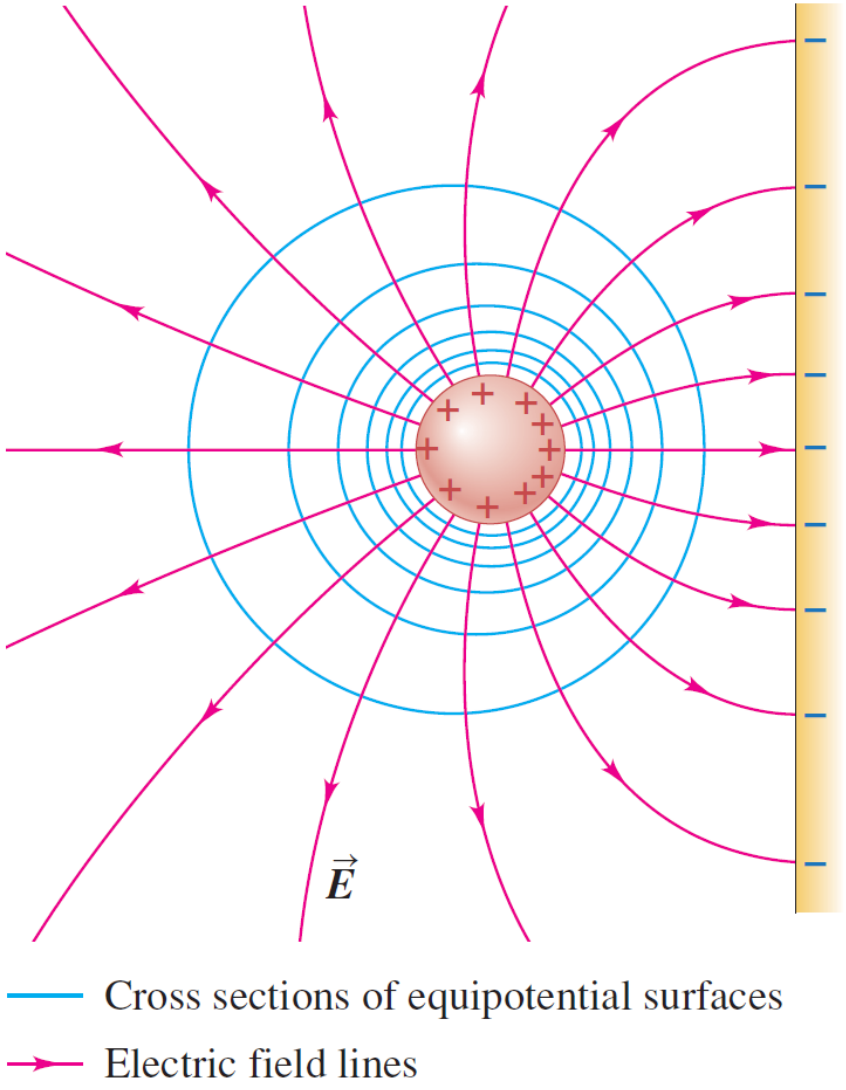
23.23 Cross sections of equipotential surfaces (blue lines) and electric field lines (red lines) for assemblies of point charges. There are equal potential differences between adjacent surfaces. Compare these diagrams to those in Fig. 21.28, which showed only the electric field lines.



23.4 Equipotential Surfaces

When charges are at rest, a conducting surface is always an equipotential surface.

Field lines are perpendicular to a conducting surface.

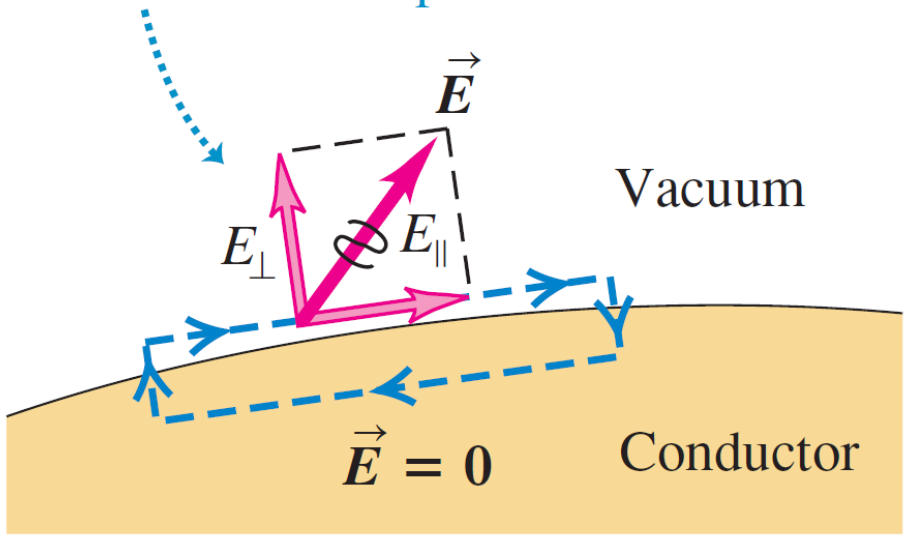


23.4 Equipotential Surfaces

At all points on the surface of a conductor, the electric field must be perpendicular to the surface. If $\vec{E}$ had a tangential component, a net amount of work would be done on a test charge by moving it around a loop as shown here—which is impossible because the electric force is conservative.

An impossible electric field

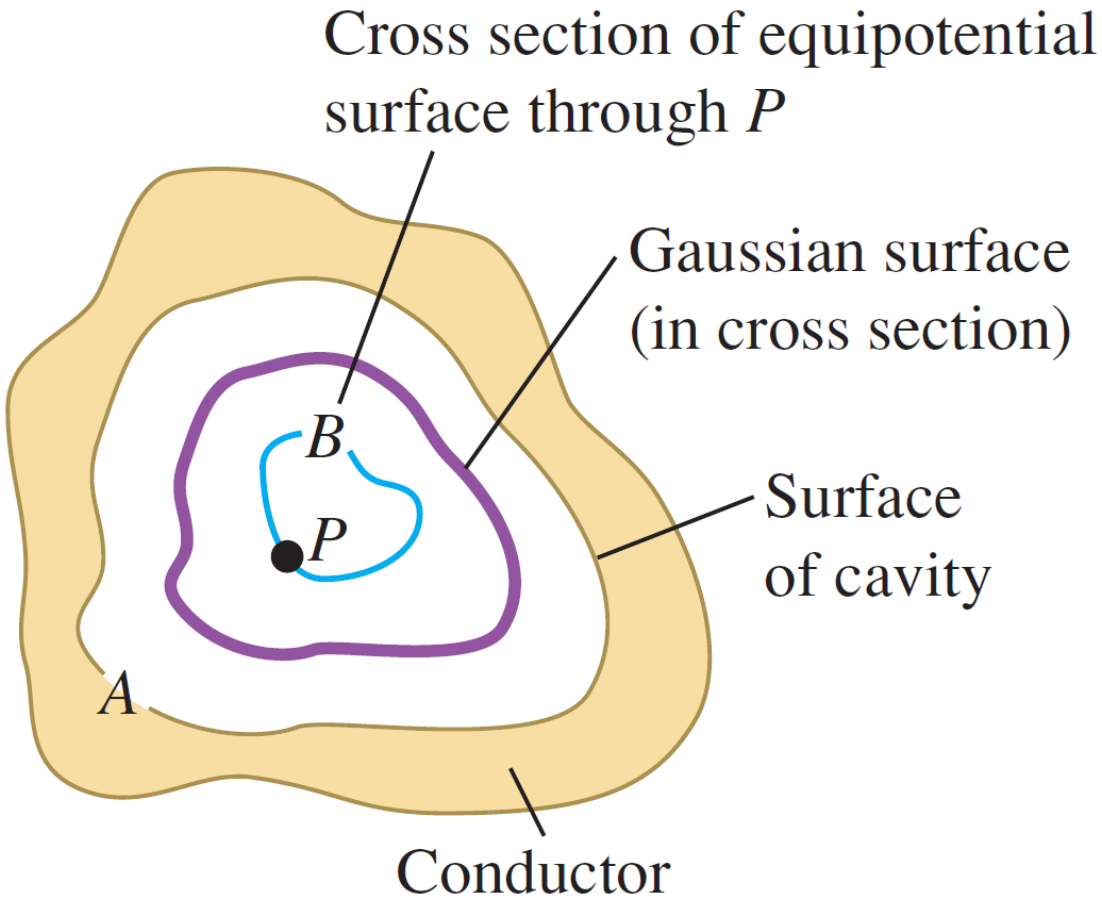
If the electric field just outside a conductor had a tangential component $E_{\parallel}$, a charge could move in a loop with net work done.



23.4 Equipotential Surfaces

A cavity in a conductor.

If the cavity contains no charge, every point in the cavity is at the same potential, the electric field is zero everywhere in the cavity, and there is no charge anywhere on the surface of the cavity.



Chapter 23.5

Potential Gradient

23.5 Potential Gradient

- Electric field, can be represented by three components written as follows:

$$E_x = -\frac{\partial V}{\partial x} \qquad E_y = -\frac{\partial V}{\partial y} \qquad E_z = -\frac{\partial V}{\partial z} \qquad (23.19)$$

(components of $\vec{E}$ in terms of V)

- In vector form, the equation is

$$\vec{E} = -\left(\hat{\imath}\frac{\partial V}{\partial x} + \hat{\jmath}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}\right) \quad \left(\vec{E}\ in\ terms\ of\ V\right) \quad (23.20)$$

Example 23.13 Potential and field of a point charge

From Eq. (23.14) the potential at a radial distance r from a point charge q is $V = q/4\pi\varepsilon_0 r$.

Find the vector electric field from this expression for V.

Identify and Set Up

By symmetry, the electric field has only a radial component E_r; we use Eq. (23.23) to find this component.

Execute

From Eq. (23.23),

$$E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r}\left(\frac{1}{4\pi\varepsilon_0}\frac{q}{r}\right) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

So the vector electric field is

$$\vec{\boldsymbol{E}} = \hat{\boldsymbol{r}}E_r = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$

Evaluate

Our result agrees with Eq.(21.7), as it must.

An alternative approach is to ignore the radial symmetry, write the radial distance as $r = \sqrt{x^2 + y^2 + z^2}$, and take the derivatives of with respect to x, and as in Eq. (23.20). We find

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x}\left(\frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{x^2 + y^2 + z^2}}\right)$$

$$= -\frac{1}{4\pi\varepsilon_0}\frac{qx}{\left(\sqrt{x^2 + y^2 + z^2}\right)^{\frac{3}{2}}} = -\frac{qx}{4\pi\varepsilon_0}\frac{1}{r^3}$$

Evaluate

And similarly

$$\frac{\partial V}{\partial y} = -\frac{qy}{4\pi\varepsilon_0 r^3} \qquad \frac{\partial V}{\partial z} = -\frac{qz}{4\pi\varepsilon_0 r^3}$$

From Eq. (23.30), the electric field is

$$\vec{\boldsymbol{E}} = -\left[\hat{\boldsymbol{\imath}}\left(-\frac{qx}{4\pi\varepsilon_0 r^3}\right) + \hat{\boldsymbol{\jmath}}\left(-\frac{qy}{4\pi\varepsilon_0 r^3}\right) + \hat{\boldsymbol{k}}\left(-\frac{qz}{4\pi\varepsilon_0 r^3}\right)\right]$$

$$= \frac{q}{4\pi\varepsilon_0 r^2}\left(\frac{x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}}{r}\right) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$

Evaluate

This approach gives us the same answer, but with a bit more effort.

Clearly, it's best to exploit the symmetry of the charge distribution whenever possible.

Example 23.14 Potential and field outside a charged conducting cylinder

In Example 23.10 (Section 23.31) we found that the potential outside a charged conducting cylinder with radius R and charge per unit length λ is

$$V = \frac{\lambda}{2\pi\varepsilon_0} ln \frac{R}{r} = \frac{\lambda}{2\pi\varepsilon_0} (\ln R - lnr)$$

Find the components of the electric field outside the cylinder.

Identify and Set Up

As in Example 23.13, the symmetry tells us that $\vec{E}$ has only a radial component E_r, so we again use Eq. (23.23).

Execute

The derivative of $\ln R$ (a constant) is zero, and the derivative of $\ln r$ is $1/r$, so

$$E_r = -\frac{\partial V}{\partial r} = \frac{\lambda}{2\pi\varepsilon_0}\frac{\partial(lnr)}{\partial r} = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$$

Evaluate

This is the same expression for E_r that we found in Example 22.6 (Section 22.4).

Example 23.15 Potential and field of a ring of charge

In Example 23.11 (Section 23.3) we found that for a ring of charge with radius a and total charge Q, the potential at a point P on the ring axis a distance x from the center is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{x^2 + a^2}}$$

Find the electric field at P.

Identify and Set Up

From the symmetry of the charge distribution shown in Fig. 23.20, the electric field along the symmetry axis of the ring can have only an x-component.

We find it using the first of Eqn. (23.19).

Execute

The x-component of the electric field is

$$E_x = -\frac{\partial V}{\partial x} = \frac{1}{4\pi\varepsilon_0}\frac{Qx}{(x^2+a^2)^{3/2}}$$

Evaluate

This agrees with the result that we obtained in Example 21.10 (Section 21.5).

Concept Summary

Electric potential energy

- The electric force caused by any collection of charges at rest is a conservative force.

- The work W done by the electric force on a charged particle moving in an electric field can be represented by the change in a potential-energy function U.

$$W_{a\to b} = U_a - U_b = -(U_b - U_a) = -\Delta U \qquad (23.2)$$

(work done by a conservative force)

Electric potential energy

- The electric potential energy for two point charges q and q_0 depends on their separation r.

$$U = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r} \qquad (23.9)$$

(electric potential energy of two point charges q and q_0)

Electric potential energy

- The electric potential energy for a charge q_0 in the presence of charges q_1, q_2, q_3 depends on the distance from q_0 to each of these other charges.

$$U = \frac{q_0}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \cdots\right)$$

$$= \frac{q_0}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i} \qquad (23.10)$$

(point charge q_0 and collection of charge q_i)

Electric potential

- Potential, denoted by V, is potential energy per unit charge.
- The potential difference between two points equals the amount of work per charge that would be required to move a positive test charge between those points.

$$V = \frac{U}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \qquad (23.14)$$

(potential due to a point charge)

Electric potential

- The potential V due to a quantity of charge can be calculated by summing (if the charge is a collection of point charges) or by integrating (if the charge is a distribution).

$$V = \frac{U}{q_0} = \frac{1}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i} \qquad (23.15)$$

(potential due to a collection of point charges)

Electric potential

- The potential difference between two points a and b, also called the potential of a with respect to b, is given by the line integral of $\vec{\boldsymbol{E}}$.

$$V = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r} \qquad (23.16)$$

(potential due to a continuous distribution of charge)

Electric potential

- The potential at a given point can be found by first finding $\vec{E}$ and then carrying out this integral.

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E \cos\phi \, dl \qquad (23.17)$$

(potential difference as an integral $\vec{E}$)

Equipotential surfaces

- An equipotential surface is a surface on which the potential has the same value at each point.
- At a point where a field line crosses a equipotential surface, the two are perpendicular.
- When all charges are at rest, the surface of a conductor is always an equipotential surface and all points in the interior of a conductor are at the same potential.

Equipotential surfaces

- When a cavity within a conductor contains no charge, the entire cavity is an equipotential region and there is no surface charge anywhere on the surface of the cavity.

Finding electric field from electric potential

- If the potential V is known as a function of the coordinates x, y, and z, the components of electric field $\vec{E}$ at any point are given by partial derivatives of V.

$$E_x = -\frac{\partial V}{\partial x} \qquad E_y = -\frac{\partial V}{\partial y} \qquad E_z = -\frac{\partial V}{\partial z} \tag{23.19}$$

(components of $\vec{E}$ in terms of V)

$$\vec{E} = -\left(\hat{\imath}\frac{\partial V}{\partial x} + \hat{\jmath}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}\right) \quad \left(\vec{E}\ in\ terms\ of\ V\right) \tag{23.20}$$

Homework

23.20

23.20 Two point charges of equal magnitude Q are held a distance d apart. Consider only points on the line passing through both charges.

(a) If the two charges have the same sign, find the location of all points (if there are any) at which (i) the potential (relative to infinity) is zero (is the electric field zero at these points?), and (ii) the electric field is zero (is the potential zero at these points?).

(b) Repeat part (a) for two points charges having opposite signs.